

Investigating Postoperative Hypoxemia Using the Medical Informatics Operating Room Vitals and Events Repository (MOVER)

Authors¹

¹Dalla Lana School of Public Health, Biostatistics Division, University of Toronto, Toronto, Canada

Abstract

Introduction & Literature Review

- Postoperative hypoxemia is a frequent complication from anesthetic¹
- The MOVER EPIC dataset includes records from 64,354 surgeries between 2017 and 2022² and can be used to study this issue
- Prior research has assessed how postoperative hypoxemia is affected by different variables, but these models only contained up to four laboratory test results^{3,4,5}
- To fill this gap characterized by few laboratory test results, this work modelled postoperative hypoxemia probability using logistic regression with up to ten preoperative laboratory test results and three demographic variables as covariates
- The objective was to detect significant predictors of postoperative hypoxemia and determine the accuracy of predicting its probability with just these variables

Methodology (The Pipeline)

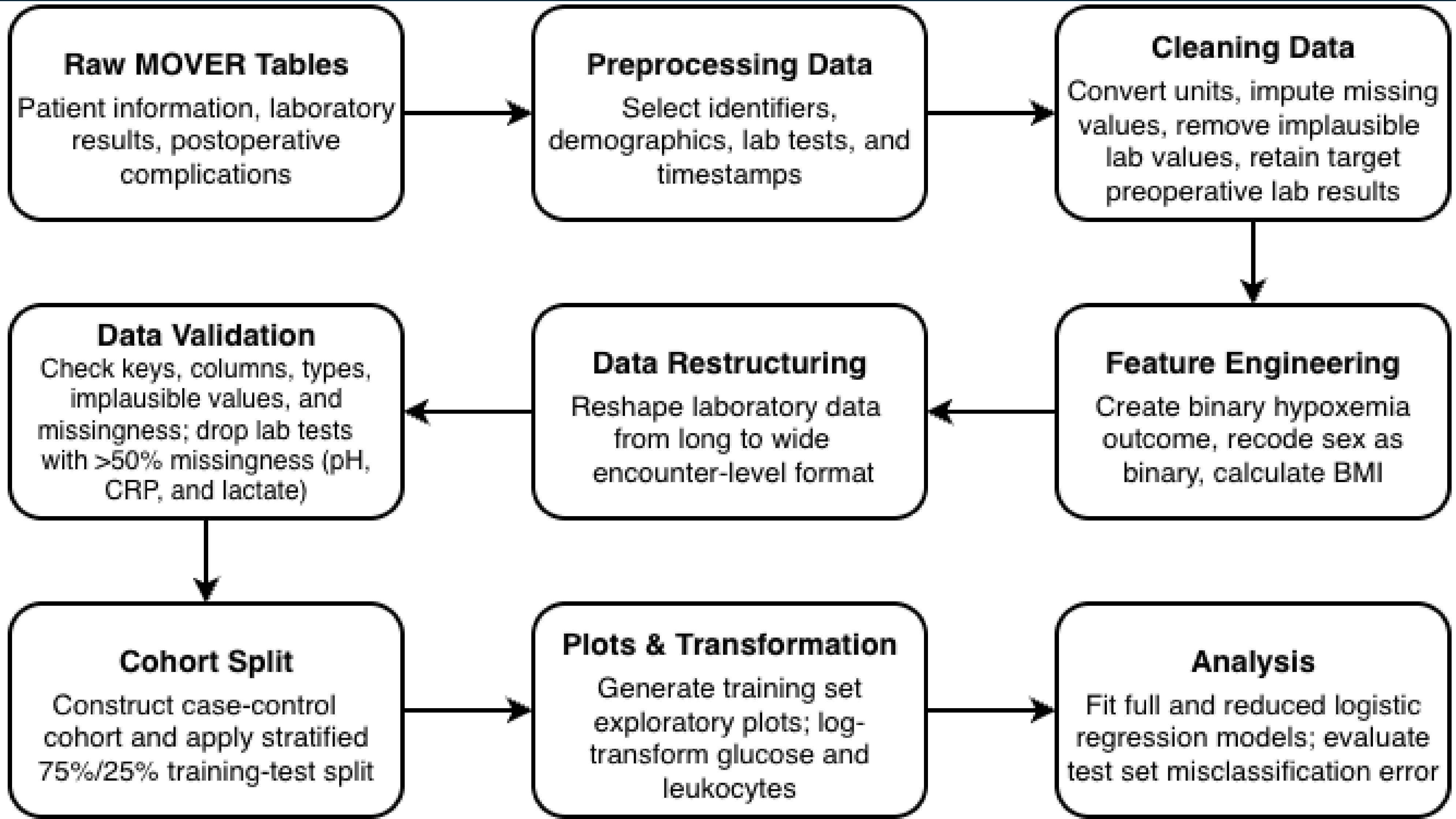


Figure 1. Flow chart of data wrangling architecture

Key Findings

Variable	Coefficient Estimate	P-value
Intercept	-6.576	0.237
Age	0.004	0.660
Sex: Female	-1.426	0.001
Body mass index	0.051	0.040
Carbon dioxide	0.015	0.716
log(Glucose)	0.786	0.114
Hematocrit	0.094	0.557
Hemoglobin	-0.521	0.264
log(Leukocytes)	0.257	0.485
Potassium	0.117	0.717
Sodium	0.014	0.650

Table 1. Full logistic regression model results for postoperative hypoxemia

- On the test set, the full model had a misclassification error rate of 0.293 (29.3%), while the reduced model, which included only the predictors that were significant in the full model, had a misclassification error rate of 0.28 (28.0%). Thus, the reduced model showed slightly better predictive performance than the full model in this sample.

- Table 1 presents the full logistic regression model results. In the full model, sex and body mass index were the only predictors significant at the 0.05 level, with female sex associated with lower odds of postoperative hypoxemia and higher body mass index associated with higher odds.

Conclusion & Impact

Key Takeaway: Male patients with high BMI and high glucose may benefit from closer postoperative monitoring for early detection and timely intervention of hypoxemia.

Key Risk Factors

- Higher BMI** elevates odds of postoperative hypoxemia , aligning with established literature⁴
- Higher glucose level** were significantly associated with increased odds of hypoxemia
- Male** patients have higher odds of hypoxemia, which contradicts previous findings that showed being female as a risk factor⁴

Model Performance and Limitation

- The full model achieved a lower test misclassification rate (37%) than the reduced model (43%), suggesting that including a broader set of lab test results is necessary to capture the complexity of hypoxemia
- Only [percentage]% of patients reported postoperative hypoxemia, likely contributing to the relatively high misclassification rate

Future Directions

- Employing oversampling or re-weighting techniques to better account for imbalance and improve predictive accuracy
- Incorporating a wider variety of laboratory tests to strengthen predictive power
- Exploring how hypoxemia interacts with other concurrent postoperative complications

References

- ¹ Yosef Belay Bizuneh, Mulualem Endeshaw Zeleke, Ashebir Mamay Gebru, Destaw Fetene Teshome, Fantahun Ayenew Mekonnen, Dessie Abebaw Angaw, and Amare Belete Getahun. 2025. Global prevalence of postoperative hypoxemia among adult and pediatric surgical patients: a systematic review and meta-analysis. *BMC Anesthesiology* 25, Article 284 (June 2025), 10 pages. doi:10.1186/s12871-025-03146-3
- ² Muntaha Samad, Mirana Angel, Joseph Rinehart, Yuzo Kanomata, Pierre Baldi, and Maxime Cannesson. 2023. Medical Informatics Operating Room Vitals and Events Repository (MOVER): a public-access operating room database. *JAMIA Open* 6, 4, Article ooad084 (Oct. 2023), 8 pages. doi:10.1093/jamiaopen/ooad084
- ³ Huiqing Ge, Ye Jiang, Qijun Jin, Linjun Wan, Ximing Qian, and Zhongheng Zhang. 2018. Nomogram for the prediction of postoperative hypoxemia in patients with acute aortic dissection. *BMC Anesthesiology* 18, Article 146 (Oct. 2018), 9 pages. doi:10.1186/s12871-018-0612-7
- ⁴ Ming Gong, Zining Wu, Shijun Xu, Lei Li, Xiaolong Wang, Xinliang Guan, and Hongjia Zhang. 2019. Increased risk for the development of postoperative severe hypoxemia in obese women with acute type a aortic dissection. *Journal of Cardiothoracic Surgery* 14, Article 81 (April 2019), 9 pages. doi:10.1186/s13019-019-0888-9
- ⁵ Debas Yaregal Melesse, Zewditu Abdissa Denu, Habtamu Getinet Kassahun, and Abatneh Feleke Agegnehu. 2020. The incidence of early post-operative hypoxemia and its contributing factors among patients underwent operation under anesthesia at University of Gondar comprehensive and specialized referral hospital, Gondar, North West Ethiopia, 2018. A prospective observational study. *International Journal of Surgery Open* 22 (2020), 38–46. doi:10.1016/j.ijso.2019.11.011